Focal Contrastive Learning for Palm Vein Authentication

Yuer Ma, Haizhen Huang, Dacan Luo, Shifeng Zhang, Wenxiong Kang, *Member, IEEE*, and Di Xie

***Abstract*— With the advancement of deep learning technology, the palm vein authentication technology based on convolutional neural network (CNN) has been greatly developed. Among many methods based on CNN, contrastive learning stands out for its excellent performance in various visual tasks. It enables machines to better understand how objects differ from each other within a given category, which is well suited to the task of fine-grained identification. This inspires us to apply contrastive learning to the palm vein authentication task. In response, we propose a focal contrastive palm vein network (FCPVN) for palm vein authentication. First, label information was introduced into self-supervised contrastive learning to create a palm vein authentication paradigm based on supervised contrastive learning. On this basis, we design a novel loss named focal contrastive loss, which employs a hard example mining strategy by introducing two factors to make the model pay more attention to hard examples. Extensive experiments on five public palm vein databases show that FCPVN has competitive performance compared to existing palm vein authentication methods.**

***Index Terms*— Authentication, biometric technology, contrastive learning, hard sample mining, palm vein.**

I. Introduction

BIOMETRIC technology has been studied extensively with increasing attention to information security. Face, fingerprint, palm vein, finger vein, gait, and voice are widely used biometric features for identity authentication [1], [2], [3], [4], [5], [6], [7], [8], [9] nowadays. Among them, palm vein is excellent for authentication tasks due to its high security and user-friendliness, which has attracted much attention from many researchers [10], [11], [12], [13]. Moreover, with the great development of deep learning, which shows massive energy in palm vein authentication tasks, more convolutional neural network (CNN)-based methods have been proposed with better performance.

Recently, contrastive learning has been widely applied in unsupervised learning for various computer vision (CV) tasks [14], [15], [16], [17], [18]. It helps the model to learn the strong discriminative features even without labeled data. The model is able to distinguish positive and negative cases of each example by increasing the similarity of positive example pairs and decreasing the similarity of negative example pairs in the representation space. Due to its excellent ability to extract discriminative features, contrastive learning has been gradually applied to supervised settings [19], [20], [21]. For fine-grained authentication tasks, such as biometric authentication, the key problem for researchers is how to effectively distinguish between positive and negative example pairs. This means that contrastive learning can be an ideal solution.

Most CNN-based palm vein authentication methods use CNNs as the feature extractor, and the similarity of the features is ultimately calculated for identity authentication. By pulling the feature embeddings of different images of the same palm close together and pushing the feature embeddings of the images of different palms apart, metric loss functions can help the model learn highly discriminative visual representations. Thus, many works have designed metric loss functions [22], [23], [24] to monitor the model training and constrain the distances between examples. However, instead of taking into account how discriminable different example pairs are, these works perform gradient backpropagation of each example pair in a fair manner. Since most examples tend to have good discrimination in the feature space, there is little useful information in the later training process, resulting in the model falling into the local optimal solution. Furthermore, the hard examples are still not embedded in the correct location in the feature space, and the authentication performance of the model is limited. Therefore, to improve the authentication performance of the model, how to design a well-performing metric function to exploit the effective information of hard examples is still a problem to be solved.

As considered above, we draw on the idea of supervised contrastive learning and propose a focal contrastive palm vein network (FCPVN). A novel loss function named focal contrastive loss is designed, which makes full use of example information to prevent the network from falling into a locally optimal solution. Each example pair is given a different weight to participate in the calculation of the loss function. Therefore, in the gradient backpropagation process, the network pays

Manuscript received 19 February 2023; revised 25 June 2023; accepted 27 July 2023. Date of publication 21 August 2023; date of current version 18 September 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 61976095 and in part by the Natural Science Foundation of Guangdong Province, China, under Grant 2022A1515010114. The Associate Editor coordinating the review process was Dr. Adam G. Polak. *(Yuer Ma and Haizhen Huang contributed equally to this work.) (Corresponding authors: Wenxiong Kang; Di Xie.)*

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Digital Object Identifier 10.1109/TIM.2023.3304689

different attention to learn the features of each example pair according to their learning difficulty. Specifically, the network pays more attention to learning the features of difficult examples.

In general, our contributions are summarized as follows.

1) We develop an FCPVN that can effectively learn strong discriminative features and discriminate between positive and negative palm vein examples. To the best of our knowledge, this is the first work to implement supervised contrastive learning for palm vein authentication tasks.
2) We employ a hard mining strategy to prevent network overfitting in the training process. This strategy allows the network to focus on hard palm vein examples close to the decision boundary, preventing premature convergence, and avoiding over-reliance on simple palm vein examples.
3) We propose a method that maintains consistent computational complexity and efficient training, regardless of the size of the database.

II. Related Works

A. Palm Vein Authentication

Palm vein authentication process involves several steps: image acquisition, image preprocessing, feature extraction, and feature matching. The region of interest (ROI) of the collected palm images is intercepted to focus on the vein area. In the image preprocessing stage, image enhancement techniques, such as color jitter and random rotation, are usually applied to simulate lighting and posture changes in the actual application scene. The features in the enhanced ROI images can then be extracted. In the research of palm vein authentication, many texture information extraction methods have been proposed. These methods can be broadly divided into two categories: 1) palm vein vessel geometry-based method: directly using the detailed vein information in the image to describe features and 2) ROI holistic-based method: taking the palm vein image as a whole and using image information to describe features.

The vascular geometry-based method extracts the directional and positional information of ridges, lines, and feature points through edge detection algorithms [25], [26], [27], [28]. However, these methods are sensitive to the rotation, translation of the palm posture, and scaling of the palm vein images.

The ROI holistic-based method can be further divided into three categories.

1) *Texture-Based Methods [29], [30], [31]:* The texture features in the images are statistically analyzed and the histogram distribution of the palm vein images is described.
2) *Local Invariant-Based Methods [10], [32], [33]:* They find the key points of the palm vein images and perform key points matching during identification.
3) *Subspace-Based Methods [34], [35], [36]:* They reflect the input images into a subspace constructed from a training set to simplify the data structure of the variables.

Traditional palm vein authentication methods have poor generalization and algorithm robustness. With the development of deep learning method and its excellent performance and robustness, some scholars started to explore realizing palm vein authentication based on deep learning methods. Hassan and Abdulrazzaq [13] designed a six-layer CNN for the extraction of palm vein features. Jia et al. [37] used a pretrained network for feature extraction and then performed a spatial pyramid pooling operation. A linear support vector machine (SVM) is connected behind the network for the classification task. As fine-grained authentication tasks require models with strong feature discrimination, many researchers have further improved the network and loss function. They explore the design of a strong model that can discriminate between positive and negative example pairs and effectively perform identity authentication. Thapar et al. [22] proposed PVSNet, which realizes the end-to-end palm vein authentication. It uses a network to locate the ROI of the palm vein image and then used an encoder–decoder network to extract features. Finally, a Siamese network is trained for feature matching by using triplet loss [38]. Ou et al. [23] used a cosine margin softmax loss [39] and a triplet loss to support the model training.

B. Contrastive Learning

Self-supervised learning has been successfully applied in natural language processing (NLP) and CV. Also, contrastive learning is an important branch of self-supervised learning. In contrastive learning, the representation learned by the model can be able to distinguish positive and negative cases of each example by increasing the similarity of positive example pairs and decreasing the similarity of negative example pairs in the representation space, respectively. Moco [16] used a memory bank to store a queue of keys and proposed a momentum update encoder method to balance the computational cost and dictionary consistency. SimCLR [14] discards the memory bank but set a larger batch size to increase the number of negative examples, and it adds a nonlinear mapping layer after the decoder. Mocov2 [17] also adds a nonlinear mapping layer, but it changes the data enhancement mode and reduces the batch size. Its accuracy exceeds that of SIMCLR. SwAV [15] changes the idea that more negative examples can improve the model performance, and it clusters the examples that lead to similar examples. In BYOL [18], the parameters of the online branch are updated as the network is trained using gradient descent methods, but the parameters of the target branch are updated using momentum update methods. SimSiam [40] continues BYOL's idea and believes that stop gradient played an important role in preventing model collapse. Therefore, the Siamese network is used for contrastive learning, and the parameters of one branch are not updated all the time. Khosla et al. [19] successfully applied the contrastive learning method combined with the image label information in supervised learning. Subsequently, a series of supervised contrastive learning works have been successively launched. Yang et al. [20] applied contrastive learning for recommendation. They propose a supervised contrastive learning framework to pretrain the user–item bipartite graph. Zhao et al. [21] employed the labels of the data to define the positive and negative examples and then pretrain the model based on the idea of contrastive learning. The model is later fine-tuned with cross-entropy loss in the semantic segmentation task. Supervised contrastive learning takes advantage of the benefits of unsupervised contrastive learning and labeled data supervision to enhance the model performance. Building upon supervised contrastive learning, our method utilizes labeled data and employs hard sample mining strategies to identify the optimal sample representation in the feature space. This approach is a novel and practical solution for enhancing model performance through representation learning.

C. Metric Loss Functions

Metric loss functions can effectively improve the performance of the model by adding extra distance between examples so that similar examples are closer together and dissimilar examples are further apart. Center loss [41] forces examples of each category to be as close as possible to the center of the category. However, it does not limit the distance between examples of different categories. Contrastive loss not only minimizes the distance between similar examples but also makes the distance between different examples as large as possible above the given threshold α. Triplet loss [38] performs triplet sampling on all examples, making the examples in the network embedding layer features, and the distance between similar examples is smaller than that between different examples. Compared to the contrastive loss [42], this is an optimization among three examples. However, the above methods do not consider the information of positive and negative example pairs in the whole batch, and the lifted structure loss considers the distance between all examples in a batch. Similarly, N-pair loss [43] also notices that the triplet loss only pays attention to one negative example at a time, so it selects a pair of positive example pairs and several negative example pairs in each batch. Compared with the triplet loss, it optimizes all negative examples at the same time. In order to facilitate targeted learning from samples, some methods have combined metric learning with hard negative mining strategies. For example, some researchers have combined the contrastive loss with the focal loss [44] to mine hard negative samples and further improve the model performance [45], [46], [47]. The proposed method differs from these in that a new loss function is designed based on the unsupervised contrastive learning idea, rather than simply modifying the contrastive loss. The focal loss is also adapted in the proposed method to assign learning weights and design a hard negative mining strategy specific to this task, rather than addressing class imbalance.

III. METHODOLOGY

This section details the algorithm called FCPVN and shows how it can be applied to palm vein authentication.

A. FCPVN Framework

Most palm vein authentication networks contain a feature extractor $\mathcal{F}$ and a classifier $\mathcal{G}$. In the training process, $\mathcal{F}$ extracts features from the inputs x and sends the embedding feature to the classifier $\mathcal{G}$

$$z_i = \frac{\exp(c_i)}{\sum_{j=1}^{K} \exp(c_j)}. \tag{1}$$

Here, $c = \mathcal{G} \cdot \mathcal{F}(x)$, and c_i denotes the ith element of the logits output c. K denotes the number of categories

$$L_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{K} y_{ij} \log(z_{ij}). \tag{2}$$

In this formula, L_{CE} denotes the cross-entropy loss, N denotes the number of samples in the dataset, K denotes the number of classes, y_{ij} denotes the probability of the jth class in the true label of the ith sample, and z_{ij} denotes the predicted probability of the ith sample belonging to the jth class.

However, as the value of K becomes larger, the burden of calculating z_{ij} and L_{CE} increases as well. Furthermore, researchers point out that the cross-entropy loss is insufficient to help the network to extract the discriminative features [48]. The reasons can be analyzed from two aspects. First, the cross-entropy loss function cannot capture similarity information between samples. It only measures the difference in probability distribution between predicted and actual categories, ignoring the fact that similar samples should be closer in feature space, and different category samples should be farther apart. This can lead to the failure of extracting discriminative features. Second, the cross-entropy loss does not consider the different contributions of feature dimensions to the classification task, resulting in the inability to learn a suitable feature representation that aligns with the geometric structure of the feature space. This further limits the model's ability to extract discriminative features.

The above two problems motivate us to construct an efficient and powerful method for palm vein authentication. In this article, a palm vein authentication network based on contrastive learning is proposed, as shown in Fig. 1.

In self-supervised contrastive learning, $\{x_k\}_{k=1,2,\ldots,N}$ examples are taken from the training set. Each example performs data enhancement twice so that there are $2N$ examples later. These $2N$ examples in a batch are denoted by $\{\tilde{x}_{2k-1}, \tilde{x}_{2k}\}_{k=1,2,\ldots,N}$. The following loss functions are commonly used to support network learning:

$$L = -\sum_{i \in K} \log \frac{\exp(z_i \cdot z_p/\tau)}{\sum_{a \in A(i)} \exp(z_i \cdot z_a/\tau)}. \tag{3}$$

The symbols z_i, z_p, and z_a denote the embedding layer features of different samples and $\cdot$ symbol denotes the inner product. Specifically, z_i represents the embedding layer features of sample x_i. z_p represents the embedding layer features of another sample that belongs to the same view as x_i. z_a represents any other sample in the sample set excluding z_i. Here, a belongs to $A(i)$, which refers to a set containing the K numbers except for i. In other words, $A(i)$ can be defined as $K \backslash i$. In self-supervised learning, z_i and z_p are called anchor and positive, respectively. Other views from different source images are called negative. In this formula, when the positive

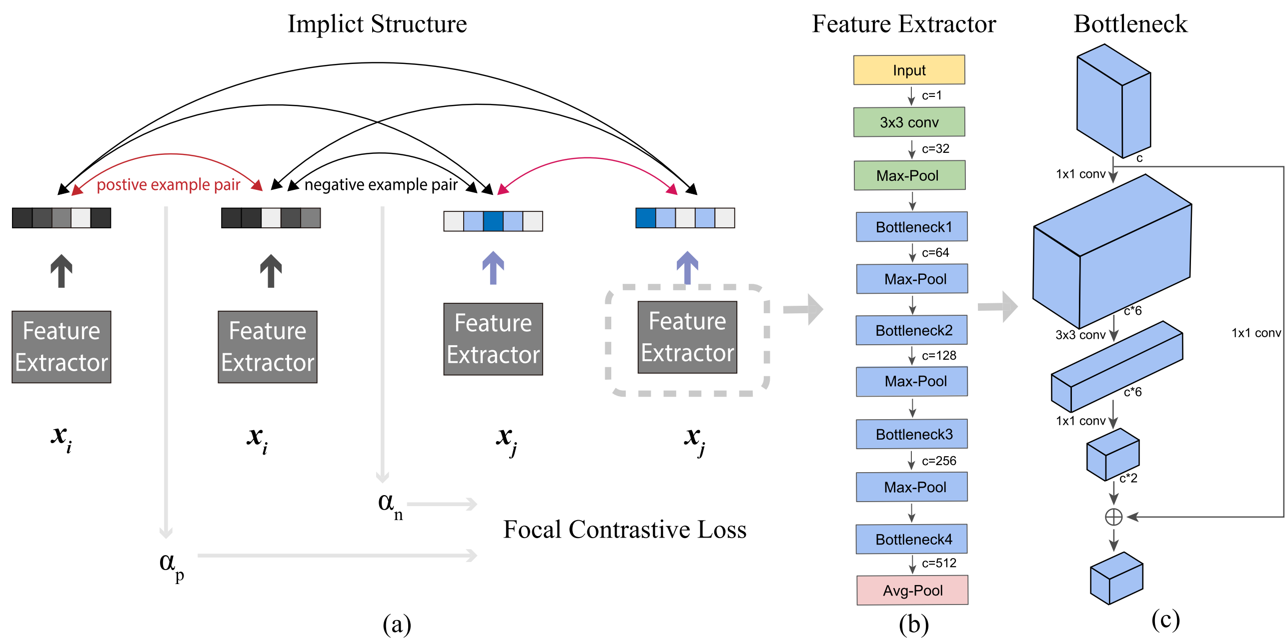


Fig. 1. Overview of FCPVN. (a) Images are fed into the network in pairs, with labels indicating whether the image pair belongs to the same identity or not. The difficulty of the example pairs is calculated in the proposed method to compute the loss function, focal contrastive loss. Here, c denotes the number of channels of the feature; (b) all of the inputs use the same feature extractor; (c) the structure of the bottleneck in the feature extractor.

sample pairs are close to each other and the negative sample pairs are farther away, the loss will gradually decrease.

The proposed method is applied in the supervised scenario. As in the process of palm vein image acquisition, tagging palm vein images is natural. A supervised learning setting is more suitable for the practical application of palm vein authentication. Meanwhile, compared to self-supervised contrastive learning, the label information can be fully utilized to build more positive and negative example pairs to increase the diversity of information learned by the network.

Considering that the input to the network in contrastive learning usually comes in pairs, the network takes the form of a Siamese network. N example pairs $\{x_i, x_j, y_{ij}\}$ are input to the network as a batch, where y_{ij} is a binary value indicating whether x_i and x_j belong to the same category. The detailed sampling strategy is described in Section III-C. Unlike most palm vein authentication methods based on CNN, the classifier and the cross-entropy loss that monitors the classification are discarded in the proposed method. Focal contrastive loss is proposed to constrain the feature of the network embedding layer that narrows the distance within a class and pushes the distance between classes apart. To make the network learn information more effectively, it is forced to focus on the hard examples. These examples contribute more knowledge to the back propagation, allowing the network to learn stronger discriminative features. More details on focal contrastive loss are described in Section III-B.

We hope to use the idea of self-supervised contrastive learning to enable the network to learn discriminative features, that is, similar examples are distributed more compactly in the feature space and dissimilar examples are distributed more dispersedly in the feature space

$$L = -\sum_{i \in K} \log \frac{\sum_{p \in P(i)} \exp(z_i \cdot z_p / \tau)}{\sum_{a \in A(i)} \exp(z_i \cdot z_a / \tau)} \tag{4}$$

where $P(i)$ denotes the set of example indices in the batch that belong to the same class as the ith example. Due to the sampling strategy, each example has at least one positive example in a batch.

In the proposed method, the distance between positive and negative sample pairs is measured, based on the idea of contrastive learning. Examples belonging to the same identity are forced to have a smaller distance, while examples belonging to different identities are forced to have a larger distance. This improves the ability of the model to distinguish between positive and negative example pairs. This approach is essentially similar to metric learning, which is a spatial mapping approach. They all involve learning a feature space in which the example is mapped into feature vectors. In this feature space, the distance between the feature vectors of similar examples is small and the distance between the feature vectors of different examples is large, allowing model to discriminate the example.

B. Focal Contrastive Loss

In object detection tasks, there is usually an imbalance in the number of positive examples and negative examples, and a large gap in the learning difficulty of the negative examples. Faced with these problems, the network always tends to correctly classify the easily classified example of a quantitatively dominant class. This is because the misclassification losses of these samples account for the majority of the total losses and dominate the gradient backpropagation. This "lazy" optimization can also cause the loss function to converge to a local minimum value. However, it goes against the expectation of better recognition performance for each class. Lin et al. [44] proposed a focal loss to solve the above problems. The loss function is shown in the following equation:

$$\mathrm{FL}(p) = -\alpha(1-p)^{\gamma}\log(p). \tag{5}$$

By adjusting the weight factor α, the weight of positive and negative examples participating in training can be modified to solve the problem of an imbalance in the number of positive and negative examples. At the same time, the adjustment factor γ can be adjusted to reduce the weight of easily classified examples participating in training. In palm vein authentication tasks, some positive example pairs may differ greatly due to ambient brightness or hand posture, while the negative example pairs may be difficult to identify whether the identity is consistent due to the blurred vein texture caused by poor photograph quality. It requires us to "emphasize" the importance of these example pairs in the network optimization process to avoid the network learning to discriminate only simple example pairs.

Therefore, the idea of adjusting the training weight of each example is integrated into contrastive learning. Inspired by focal loss, a novel loss function named focal contrastive loss is proposed in the proposed method to automatically adjust the weight of examples participating in training. The corresponding equation can be written as follows:

$$L = -\frac{1}{2N}\sum_{i\in K}\sum_{p\in P(i)}\log\frac{E_p}{\sum_{p\in P(i)}E_p + \sum_{n\in N(i)}E_n} \tag{6}$$

where

$$E_p = \omega_p \cdot \upsilon_p, E_n = \omega_n \cdot \upsilon_n \tag{7}$$

and

$$\omega_p = \left(1 - z_i \cdot z_p\right)^\alpha + \beta, \quad \omega_n = (z_i \cdot z_n)^\alpha + \beta \tag{8}$$

$$\upsilon_p = \exp\left(z_i \cdot z_p/\tau\right), \quad \upsilon_n = \exp(z_i \cdot z_n/\tau). \tag{9}$$

z_i denotes the embedding layer features of x_i. z_p and z_i refer to the embedding layer features of the examples that belong to the same category, while z_n and z_i refer to the embedding layer features of the samples that belong to different categories. z_p and z_i constitute a positive sample pair, whereas z_n and z_i constitute a negative sample pair. $P(i)$ and $N(i)$ denote example index sets that form positive and negative example pairs with example x_i, respectively. τ denotes the scalar temperature parameter.

Equation (8) contains variables υ_p and υ_n, representing the inner product between positive and negative sample pairs. The magnitude of the inner product reflects the proximity of the samples in the feature space. Equation (7) determines the weights of υ_p and υ_n in the loss function of (8). For positive sample pairs, larger weight values are assigned to pairs with greater distance between them (i.e., larger w_p values). For negative sample pairs, larger weight values are assigned to pairs with smaller distance between them (i.e., larger w_n values). α and β are exponential moderator and constant terms, respectively. Equation (6) is obtained by multiplying the inner product of each sample pair by its corresponding weight for loss calculation. E_p and E_n represent the sum of the inner products of positive and negative sample pairs, respectively. By substituting (7)–(9) into (6), the value of the loss function can be calculated.

This approach incorporates the weights of sample pairs into the loss calculation and adjusts the contribution of each sample pair to the final loss value. By doing so, the model's ability to discriminate between similar and dissimilar sample pairs can be enhanced. In (9), the positive example pairs with low similarity and the negative example pairs with high similarity are given more attention in the network training process, and they contribute more supervisory information in the loss gradient return process. Spiegl [49] also designed their network based on the idea of contrastive learning and focal loss, but did not combine the two and use contrastive learning to maintain similar features of corresponding patches in the images before and after the transformation to avoid unnecessary transformation in the style transfer. In the classification phase, the focal loss is used to solve the problem of unbalanced example categories. Different from [49], contrastive learning and focal loss are combined to transform the palm vein authentication based on multiclassification training into a binary classification task.

C. Sampling Strategy Based on Supervised Contrastive Learning

In the proposed method, the use of image labels provides greater flexibility in the construction of example pairs.

The input to the network is various example pairs. Hence, a large amount of input data can be obtained by constructing different example pairs. However, there are far more negative example pairs than positive example pairs, and constructing negative example pairs before training introduces a great deal of complexity. To overcome this problem, only positive example pairs were constructed before training. There are N positive example pairs $\{\{x_i^k, y_i^k\}, \{x_j^k, y_j^k\}, y_{ij}^k\}_{k=1,2,\ldots,N}$ input to the network as a batch. Here, $y_i^k = y_j^k = y_{ij}^k$. Then, each example in positive example pairs will also iterate through all other examples in the batch to construct positive or negative example pairs. Here, a binary variable m indicates whether the examples in an example pair belong to the same category. $m = 1$ if they are in the same category; otherwise, $m = 0$. Finally, there are $N \times N$ example pairs $\{x_i, x_j, m\}$ as input data. The two examples from each example pair are input to the two branches separately and through multiple bottlenecks and pooling layers to map to the feature subspace.

IV. Experiments

In this article, experiments are conducted on five public palm vein databases to verify the performance of the proposed method. The public databases are: CASIA MultiSpectral Palm-print Image Database V1.0 [50], PolyU Multispectral Palm-print Database [51], Tongji Contactless Palm-print&palm vein Database [52], VERA palm vein Database [53], and PUT Vein Database [54]. The experimental results show that the proposed method is very competitive. A detailed analysis of the experimental results is given in this section.

A. Databases

a) CASIA MultiSpectral Palm-print Image Database V1.0: It consists of palm images illuminated by six different wavelengths of light: 460 nm, 630 nm, 700 nm, 850 nm,

TABLE I

SUMMARY OF THE EXPERIMENTAL DATABASES. THE NUMBER OF SUBJECTS AND THE NUMBER OF EXAMPLES ARE EXPRESSED AS SUBJECTS × HANDS AND EXAMPLES × ACQUISITION SESSIONS, RESPECTIVELY. NOTED THAT, IN THE CASIA DATABASE, ONLY THE IMAGES ILLUMINATED BY THE 850-NM LIGHT WAVELENGTH WERE USED. IN THE POLYU DATABASE, ONLY THE IMAGES COLLECTED UNDER THE NIR LIGHT WERE USED

Database	subjects	examples	total images
CASIA	100×2	3×2	1,200
PolyU	250×2	6×2	6,000
VERA	50×2	5×2	1,000
PUT	50×2	4×3	1,200
Tongji	300×2	10×2	12,000

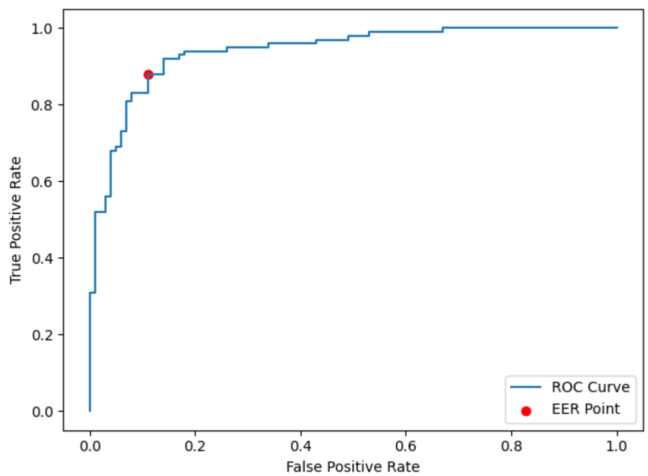


Fig. 2. Correlation between EER and ROC curve.

940 nm, and white light. In the experiments, only palm vein images illuminated by the wavelength of 850 nm are used. The palm vein images collected at this wavelength were obtained from 200 subjects three times in two sessions, for a total of 1200 images.

b) PolyU Multispectral Palm-print Database: It is collected under four different wavelengths of red, green, blue, and near-infrared light. There are 24 000 palm vein images collected from 500 subjects 12 times in two sessions under the four lights. Only the images taken under near-infrared light were used in the experiments.

c) VERA Palm Vein Database: It consists of 1000 palm vein images of 50 volunteers. Each volunteer was obtained ten palm vein images in two sessions.

d) PUT Vein Database: It collected palm vein images from 50 subjects in three different sessions for a total of 1200 palm vein images.

e) Tongji Contactless Palm Vein Database: It collected the palm vein images of 600 subjects in two different sessions, 20 times for each subject, for a total of 12 000 images. The summary of the experimental databases is shown in Table I.

B. Implementation Details

Each database mentioned above has been divided into a training set and a testing set on a 1:1 basis. The training set

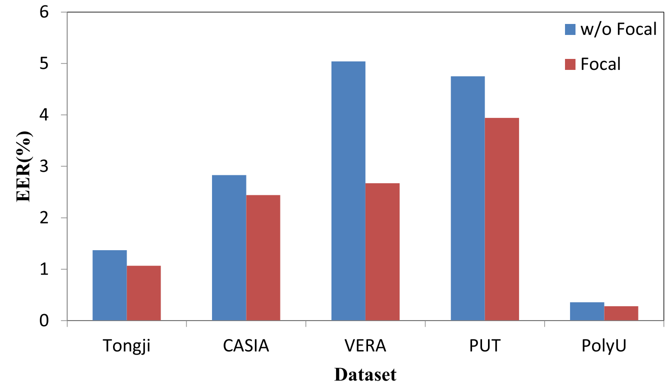


Fig. 3. Effectiveness study of hard example mining strategy: the red columns and the blue columns, respectively, indicate whether the hard example mining strategy is used for network training or not.

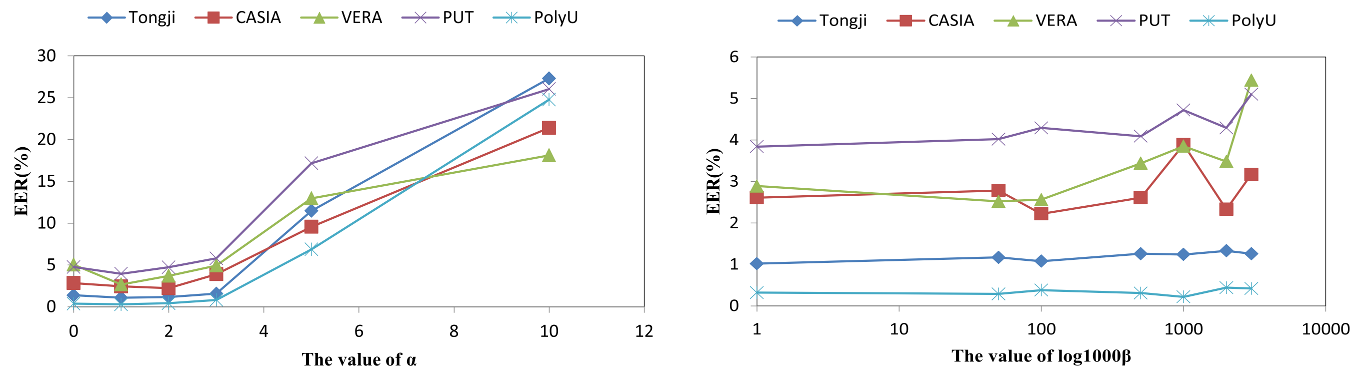
and the testing set do not have the same subject. The model is trained on the training set and the results of its performance on the testing set are reported. It is worth noting that in order to have a fair comparison with some methods based on a closed setting, the databases are split according to their setting: during the training process, all subjects are used to train the model. During the testing process, the first half of all subjects are used as registered examples and the second half of all subjects are used as probes. All registered examples are matched to probes once. Assuming that the number of subjects is M and the number of examples is N, the number of intraclass matches is $M \times N/2 \times N/2$ and the number of interclass matches is $M \times N/2 \times (M-1) \times N/2$. The input images are resized to a resolution of 224 × 224. The output of the last bottleneck layer is to embed features to calculate the focal contrastive loss for supervised training of the network. The equal error rate (EER) is employed as the performance metric. The EER is a commonly used metric in biometric systems to evaluate the accuracy of authentication. It represents the point where the two types of errors, false acceptance and false rejection, occur at the same rate. False acceptance occurs when an impostor is incorrectly accepted as a genuine user, while false rejection occurs when a genuine user is incorrectly rejected by the system. False acceptance rate (FAR) and false rejection rate (FRR) can be calculated as follows:

$$\mathrm{FAR} = \frac{\text{The number of matching pairs in false acceptance}}{\text{The number of interclass pairs}} \tag{10}$$

$$\mathrm{FRR} = \frac{\text{The number of matching pairs in false rejection}}{\text{The number of intraclass pairs}}. \tag{11}$$

In summary, the EER is a metric that quantifies the accuracy of biometric systems by measuring the balance between FAR and FRR.

The EER can be determined easily from the receiver operating characteristic (ROC) curve, which plots the true positive rate (TPR) against the false positive rate (FPR) for different decision thresholds. The EER is the point on the ROC curve where TPR equals 1 − FRR and FPR equals FAR. Biometric systems with lower EERs have better performance, achieving



(a) (b)

Fig. 4. Insensitive of hyperparameters: (a) and (b) sensitivity of α and β on five palm vein databases, respectively.

higher accuracy and a more balanced tradeoff between FAR and FRR. A simple diagram is presented in Fig. 2 for better understanding.

MobileNetv2 [55] is employed without its fully connected layer as the backbone network in all experiments. The input channel is 1 (grayscale image), and the channel of the embedding feature is 512. The Adam optimizer is used for training with an initial learning rate of 0.0025 and a momentum of 0.9 with a batch size of 64. Moreover, the cosine decay learning rate adjustment strategy is adopted. The maximum training epoch is set to 180.

C. Effectiveness Analysis

1) Effectiveness Study of Hard Example Mining Strategy: An effectiveness study is conducted to evaluate the contribution of the hard example mining strategy. The experimental results are shown in Fig. 3. "w/o Focal" means that the hard example mining strategy is not adopted in the training process (the baseline scheme), while "Focal" means that the focal contrastive loss proposed by us is used for training. It is obvious that the inclusion of the hard example mining strategy gives the network-consistent performance.

2) Sensitivity of Hyperparameters: Besides, we explore the effect of the value of the hyperparameters α and β in (8), which denote the exponential and bias in the weight calculation formula for each example. The results are shown in Fig. 4. Fig. 4(a) shows the performance from $\alpha = 0$ to $\alpha = 10$ and keeps $\beta = 0$ in all experimental databases. The trend of the lines shows that the performance remains relatively stable between $\alpha = 0$ and $\alpha = 3$. As the value of α increases, the performance gradually deteriorates. Combining the statistics across all databases in Fig. 4(a), the value of α that maximizes network performance observed is 1. Hence, the value of α is fixed at 1. Similar experiments then explore the insensitivity of β. Fig. 4(b) shows the performance from $\beta = 0$ to $\beta = 3$. To show the sensitivity of β in a more intuitive way, the abscissa value of Fig. 4(b) is set to $\log 1000\beta$. It seems that when the value of β is small, the sensitivity is small, but as the value increases, the sensitivity of some databases becomes large. The value of β is set to 0.001 in all experiments.

3) Effective of Loss Function: The backbone of FCPVN is kept, and the loss function is replaced to fairly compare the widely used methods on the identity authentication task. The results are shown in Table II. Focal Loss [44], AM-Softmax [57], Arcface [1], Sphereface [2], CIFP [58], and SCF [59] are all improved in terms of the cross-entropy loss. Cross-entropy loss makes features within classes close to each other. Focal loss is a loss function that focuses on difficult examples to improve the performance, achieved by giving more weight to misclassified examples and less weight to well-classified examples. Sphereface, Arcface, and AddMargin add constraints on the included angles between classifier weights and features, but they add constraints differently. Sphereface multiplies the cosine of the angle θ by m, Arcface adds the constant m to θ, and Addmargin adds constant m to the cosine of θ. The ultimate goal of these methods is to improve clustering within classes. SCF improves the loss convergence by introducing uncertainty into the pretrained model and changing the distribution model to a von Mises-Fisher distribution (vMF) on the hypersphere. CIFP reduces the performance bias of the authentication algorithm for different face attributes and improves the overall accuracy by introducing the false positive penalty term. The focal contrastive loss limits the intraclass and interclass spacing of example features in embedding space. Experimental results show that the loss function outperforms these methods on most databases. The differences between the proposed method and the others are analyzed in more detail by visualizing the feature distribution in Section IV-E.

D. Comparisons With Other Methods

The performance of the proposed method is compared with other methods on the five palm vein datasets. The experiments are performed in both open and closed settings, and the results are shown in Table III. The model constrains the example feature distribution in the feature space by a simple loss function and achieves excellent performance on both open and closed sets. As shown in Table III, FCPVN outperforms most methods on five palm vein databases.

TERRM [60] proposes a method for handcrafted feature extraction and feature matching based on fast Hamming distance. Ou et al. [23] generated interclass examples based on CycleGAN [62] and used the cosine margin softmax loss [39] and triplet loss to aid network training. There is a certain degree of improvement of FCPVN compared to other open-set

TABLE II

COMPARE WITH THE WIDELY USED LOSS FUNCTION IN IDENTIFICATION AUTHENTICATION FIELDS (USING EER AS AN EVALUATION INDICATOR, THE SMALLER THE EER, THE BETTER THE MODEL PERFORMANCE)

Paper	Method	Year	Tongji	CASIA	VERA	PUT	PolyU
Taigman *et al.* [56]	Softmax	2014	3.06	5.89	5.63	8.46	2.11
Lin *et al.* [44]	Focal Loss	2017	2.93	5.78	5.60	7.95	1.64
Liu *et al.* [2]	Sphereface	2017	1.13	4.00	4.00	4.29	0.38
Wang *et al.* [57]	AM-Softmax	2018	1.28	2.78	5.22	4.34	0.33
Deng *et al.* [1]	Arcface	2019	**1.01**	2.83	6.7	4.52	0.38
Xu *et al.* [58]	CIFP	2021	1.13	2.89	5.48	4.07	0.43
Li *et al.* [59]	SCF	2021	1.39	3.05	5.3	4.19	**0.27**
ours	FC Loss	2023	1.07	**2.44**	**2.67**	**3.94**	0.28

TABLE III

COMPARE WITH THE SOTA AND NEWLY METHODS (USING EER AS AN EVALUATION INDICATOR, THE SMALLER THE EER, THE BETTER THE MODEL PERFORMANCE)

Database partitioning	Paper	Method	Year	Tongji	CASIA	VERA	PUT	PolyU
open set	Cho *et al.** [60]	TERRM	2021	-	4.28	-	-	0.47
	Ou *et al.** [23]	Based on Cosineface and Triplet loss	2022	1.12	-	-	-	-
	ours	FCPVN	2023	1.07	**2.44**	**2.67**	**3.94**	0.28
closed set	Ahmad *et al.** [24]	Based on Haming Distance	2019	-	-	3.05	**0**	1.98
	Thapar *et al.** [22]	Based on triplet loss	2019	-	3.71	-	-	0.66
	Zhang *et al.** [61]	CCDFL	2022	-	-	-	-	1.02
	ours	FCPVN	2023	**0**	**0**	**0**	**0**	**0**

* denotes that results obtained from the original paper

task methods, up to 50% improvement in performance compared to these open-set authentication methods.

The wave atom transform (WAT)-based palm vein authentication method [24] generates a palm vein template by randomizing and quantizing the texture features to match the identification. PVSNet [22] uses a triplet loss and hard mining strategy for palm vein authentication in the palm vein authentication method. CCDFL [61] jointly learns multimodal features and considers the intramodal and intermodal authentication features and the quantization loss before and after encoding binary features. In the closed-set task, the performance of the model outperforms the above authentication methods and achieves the optimal performance with an EER of 0.

E. Visualization Analysis

1) Visualization of Feature Distribution: The feature distribution of the test data is visualized by using the embedding layer features of the best model for each method. The t-SNE method is used for feature visualization. The feature distributions of some popular metric learning methods mentioned above for identification authentication tasks are compared. The corresponding visualization results are shown in Fig. 5.

As shown in Fig. 5, the cross-entropy loss has the worst performance because the features within the class are not compact. As a result, examples belonging to different categories are very close. The other methods impose stronger constraints on the angles between the intraclass but are still inferior to the proposed method. Unlike these methods, the method simultaneously constrains the interclass and intraclass distances of features and shows good performance on all feature visualization images.

2) Feature Map Visualization: To better understand how the FCPVN works, the intermediate feature maps of the last bottleneck are visualized. Fig. 6 shows the activation maps of normalized features. Each column shows the original images, the activation map based on the baseline, and the activation map based on FCPVN. It can be observed that, for images from different databases, the activation maps of the scheme are more consistent than the baseline scheme. To further demonstrate the objectivity and reliability of the results, additional visualizations of more samples are provided in the Appendix.

F. Computational Complexity and Class in Loss Functions

Fig. 7 reports the floating-point operations per second (FLOPS) required to compute different loss functions in a single forward pass, as the number of classes in the dataset increases. The number of classes is set to be 10, 100, 1000, 10 000, 100 000, 100 000, and 200 000 and evaluate the FLOPS required to compute each loss function under the same network hyperparameters. During a forward propagation of the network, the computational complexity of the cross-entropy loss function increases with the number of classes in the dataset, while the designed loss function exhibits constant computational complexity regardless of the number of classes.

G. Performance in Real-World Applications

To evaluate the performance of the proposed algorithm in practical scenarios, we employ a laboratory-developed unconstrained palm vein recognition device, consisting primarily of an industrial camera, a main control board, a self-designed infrared light source circuit board for supplementary lighting, and an expansion circuit board. In practical applications, we deploy a deep neural network model using the NCNN framework. The images of the device's appearance are shown in Fig. 8.

Fig. 5. Feature distribution visualization: t-SNE of the six methods features on two databases. The features of different identities are marked in different colors.

We use an ROI-based algorithm [10] to locate the vein area in the palm images and remove excess background information. The first and second rows in Fig. 9 represent the original palm vein images acquired by the device (shown in Fig. 8) and the vein region images cropped by the ROI area extraction algorithm, respectively.

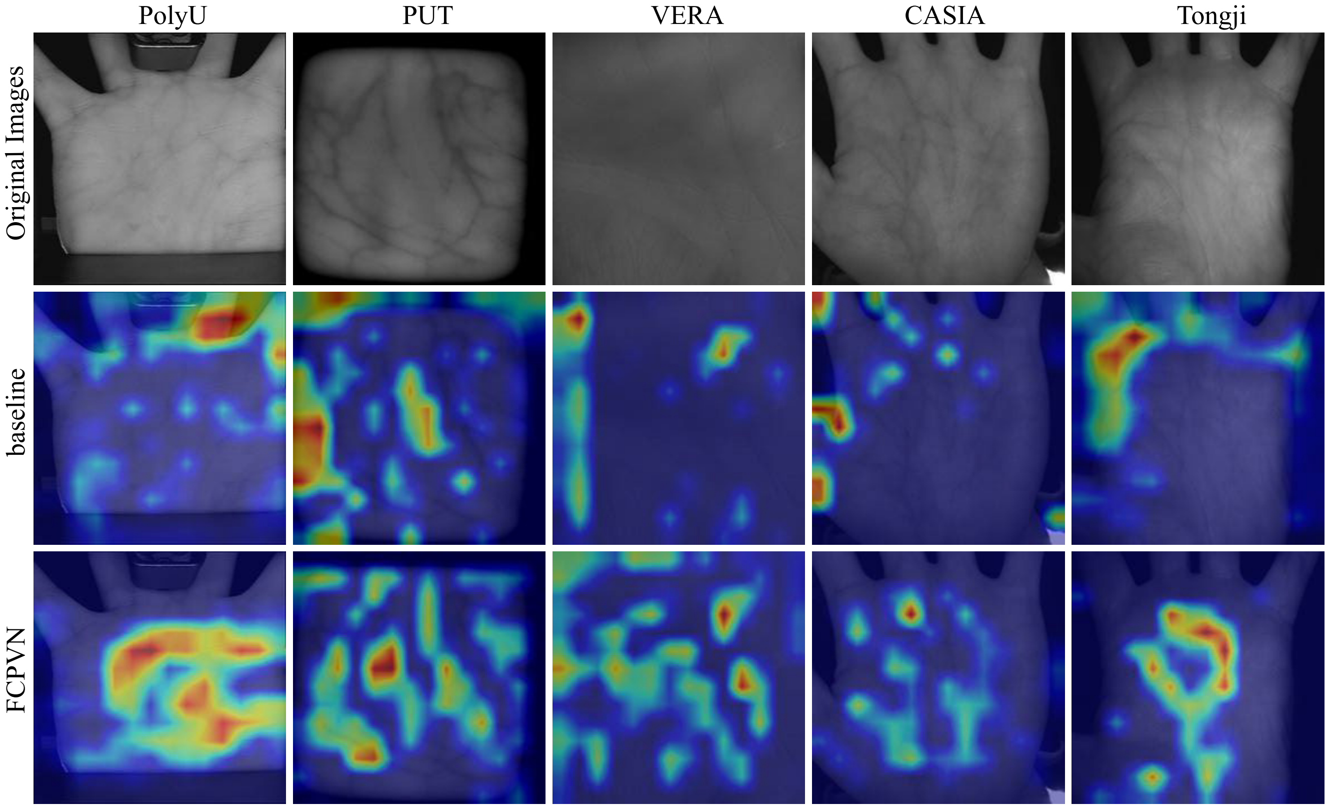


Fig. 6. Action maps visualization: the intermediate feature maps of the last bottleneck are visualized. The three columns show the original images, the activation map based on the baseline, and the activation map based on FCPVN.

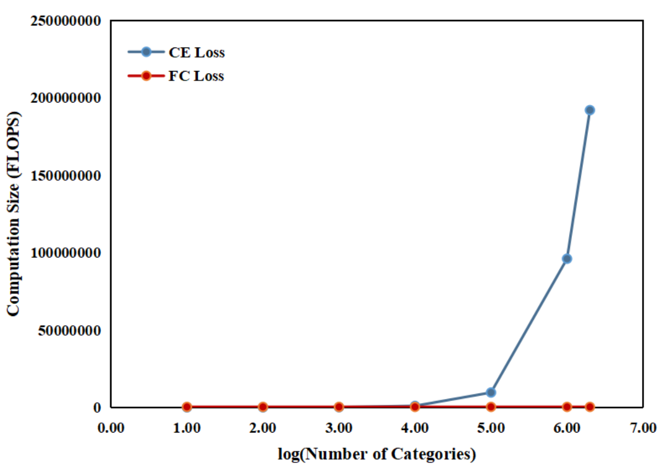


Fig. 7. Correlation between the number of categories and loss computation size.

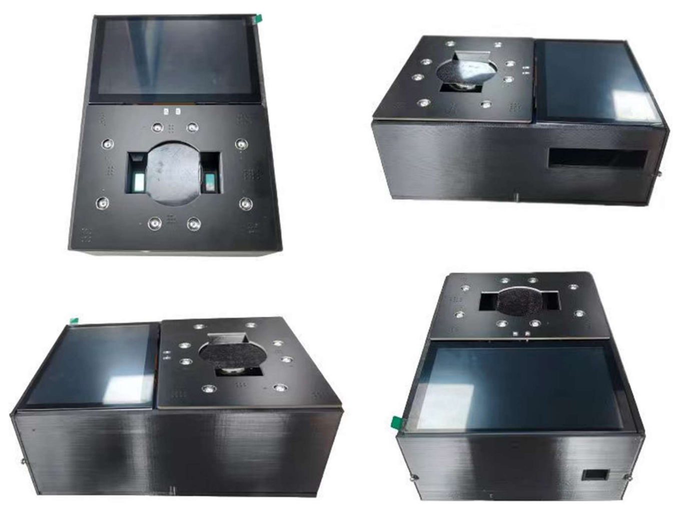


Fig. 8. Images of the device's appearance in real-world applications.

The proposed algorithm is deployed on this device along with the ROI area extraction algorithm mentioned above and tested in a real-world scenario with a sample of 40 individuals, consisting of 20 registered users and 20 unregistered users. The registered users' palm vein patterns are stored in the device and need to be correctly identified and authenticated, while the unregistered users' palm information is not registered in the device and needs to be rejected as unregistered. Each participant used the device for authentication 20 times, resulting in an FAR of around 2.5% and an FRR of 1.8%. The response time of the device is about 110 ms per authentication, indicating good accuracy and speed in real-world applications.

H. Discussion

The experimental results demonstrate the superiority of FCPVN in the task of palm vein authentication compared to existing state-of-the-art methods. Subsequently, further analysis and interpretation of the experimental results are performed.

In Section IV-C, the analysis reveals that the performance gain is attributed to the proposed Focal Contrastive Loss, which is designed to focus on the most challenging examples at decision boundaries. The hard example mining strategy

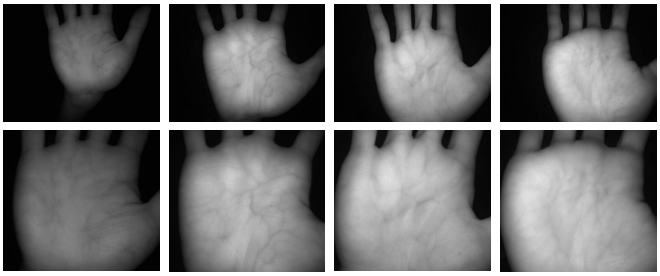


Fig. 9. Images in the first row show the palm vein images acquired from the palm vein authentication device developed in our laboratory. The images in the second row show the vein region images obtained after preprocessing the images in the first row with an ROI area extraction algorithm.

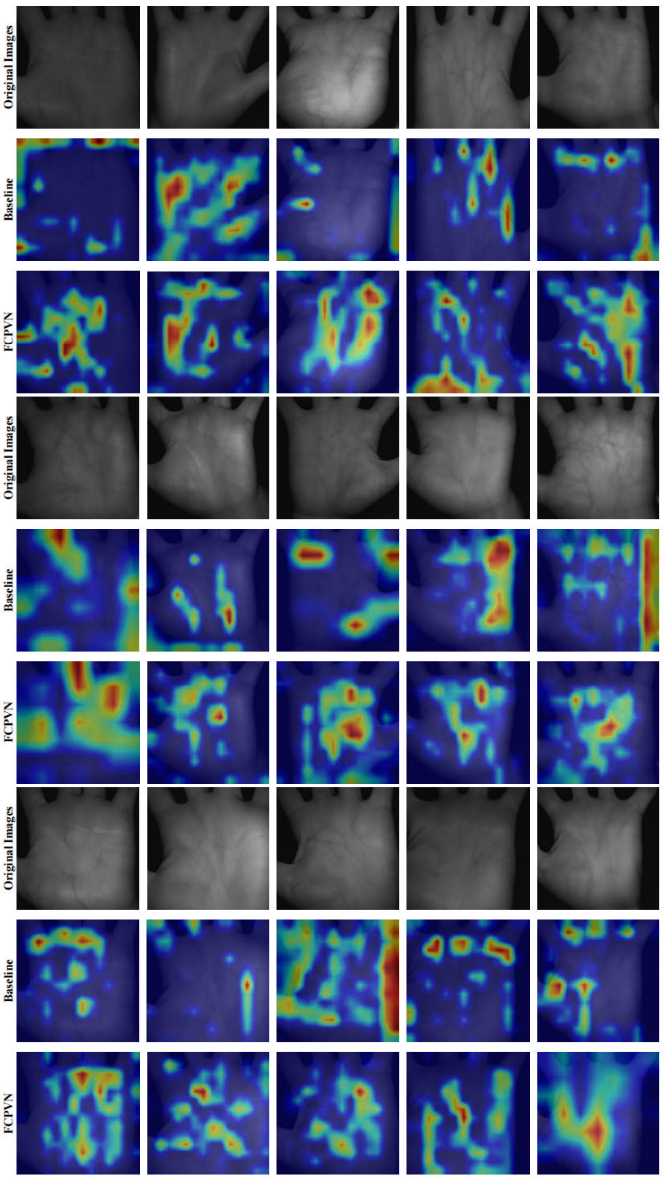


Fig. 10. Action maps visualization on the Tongji database. The text on the left of the images indicates the category of each image: original image, heatmap based on baseline, or heatmap based on FCPVN.

improves the network's ability to handle confusing examples, preventing premature convergence due to easy examples with significant intraclass and interclass spacing gaps.

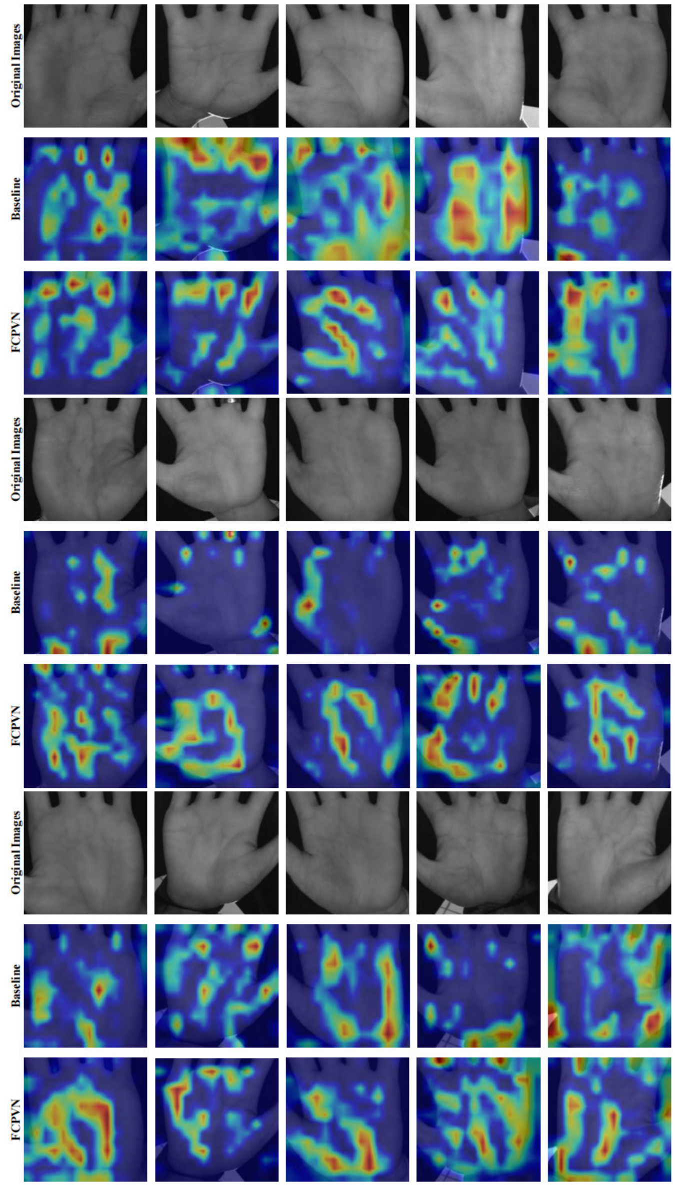


Fig. 11. Action maps visualization on the CASIA database. The text on the left of the images indicates the category of each image: original image, heatmap based on baseline, or heatmap based on FCPVN.

In Section IV-C, a comprehensive hyperparameter sensitivity analysis of focal contrastive loss is conducted. The model's sensitivity to parameter changes is reduced when α and β are small. However, if they exceed a certain threshold, model performance may decrease. Large parameter values amplify the weight of difficult examples, causing the model to prioritize their learning over that of the entire dataset. This compromises the model's ability to learn from simple examples and decrease its performance on the test set. Therefore, selecting appropriate parameter values is critical to ensure an optimal balance between the learning of difficult and simple examples.

To further analyze the working principle of focal contrastive loss, two visualization experiments are conducted in Section IV-E.

In the first paragraph, a visualization experiment is conducted on feature distributions. The softmax loss minimizes

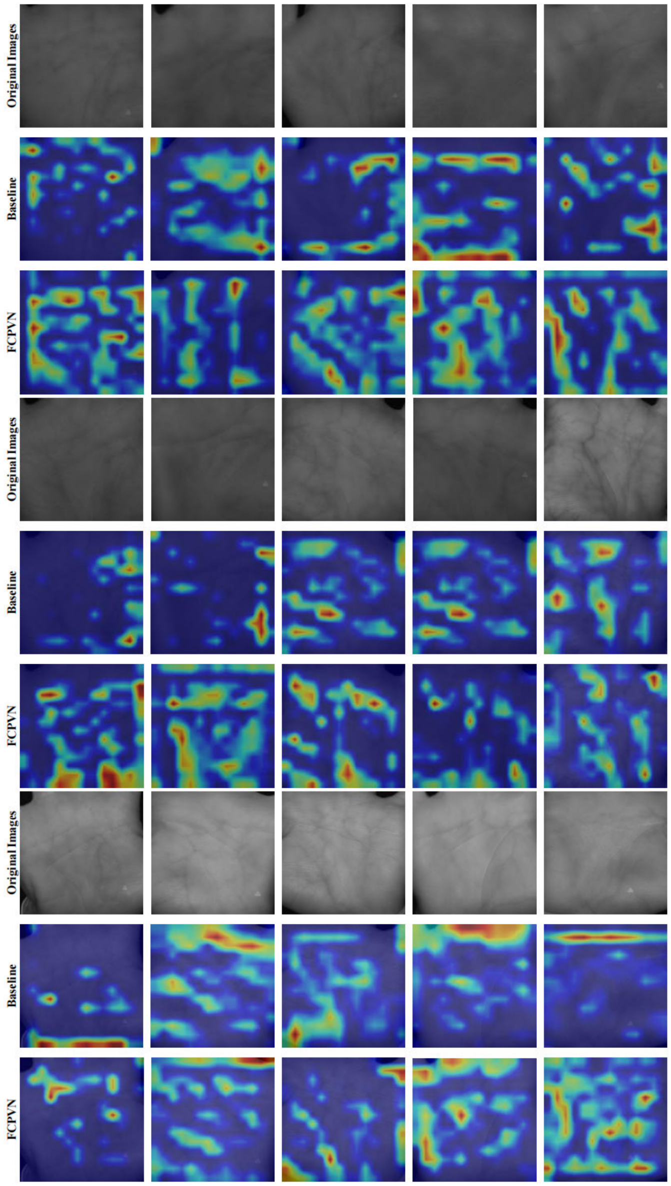


Fig. 12. Action maps visualization on VERA database. The text on the left of the images indicates the category of each image: original image, heatmap based on baseline, or heatmap based on FCPVN.

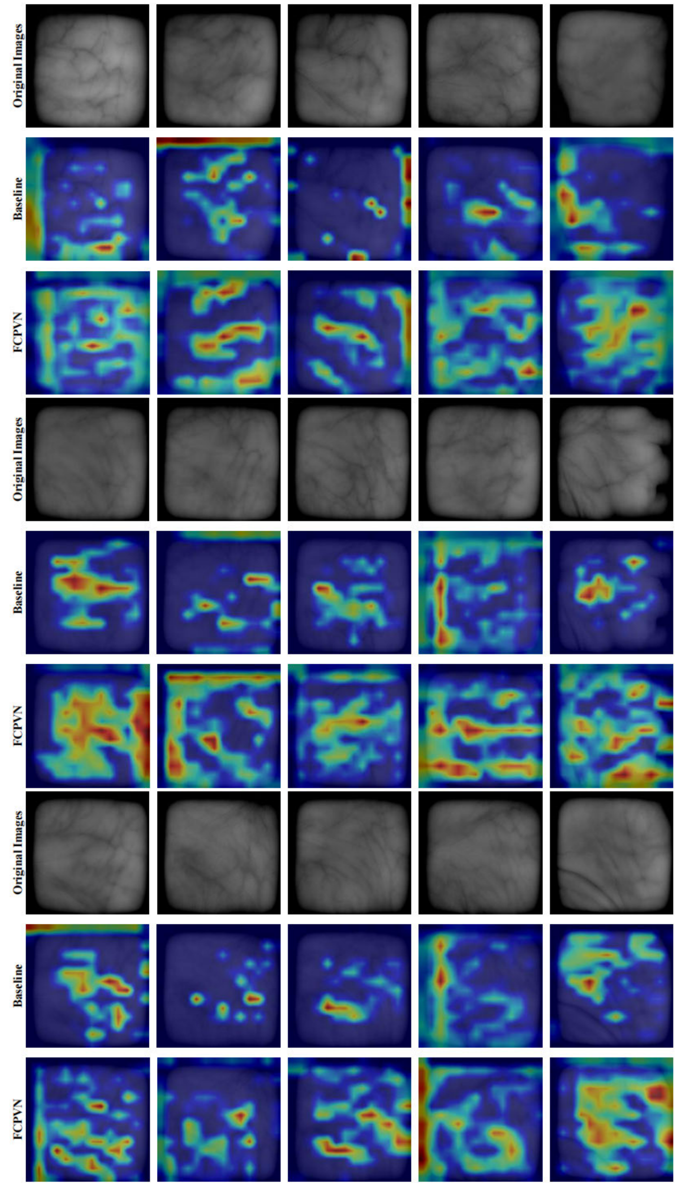


Fig. 13. Action maps visualization on the PUT database. The text on the left of the images indicates the category of each image: original image, heatmap based on baseline, or heatmap based on FCPVN.

the angle between features from the same class, but it does not impose constraints on intraclass compactness and interclass discreteness, making it unsuitable for open-set authentication tasks such as palm vein authentication. Other improved methods based on softmax are superior to softmax in feature clustering but inferior to the results. These methods rely on the compactness of intraclass features to determine the distances between classes, rather than using explicit constraints, leading to uneven feature distributions. Under the supervision of the focal contrastive loss, features are uniformly distributed in feature space, resulting in the best clustering effect in feature visualization while simultaneously ensuring intraclass compactness and interclass discreteness.

In the second paragraph, a visualization experiment is conducted on feature maps. Compared with the baseline method, the proposed method is more accurate in activating the vein region. The reason is that the baseline method is prone to misidentifying the shadow areas or contour information on the surface of the palm as vein regions, resulting in a more disorganized activation map. In contrast, the proposed method can more accurately identify and activate the true vein regions, which generates more useful embedding features and improves the accuracy of the testing process. Such improvements can help enhance the application value and reliability of palm vein recognition technology.

In Section IV-F, we compare the computational complexity of the designed loss function and the cross-entropy loss function for datasets with different numbers of classes. The experiments showed that the proposed method is computationally superior because it only requires calculating similarity scores between positive and negative samples in a single batch, while the cross-entropy loss function needs to compute the

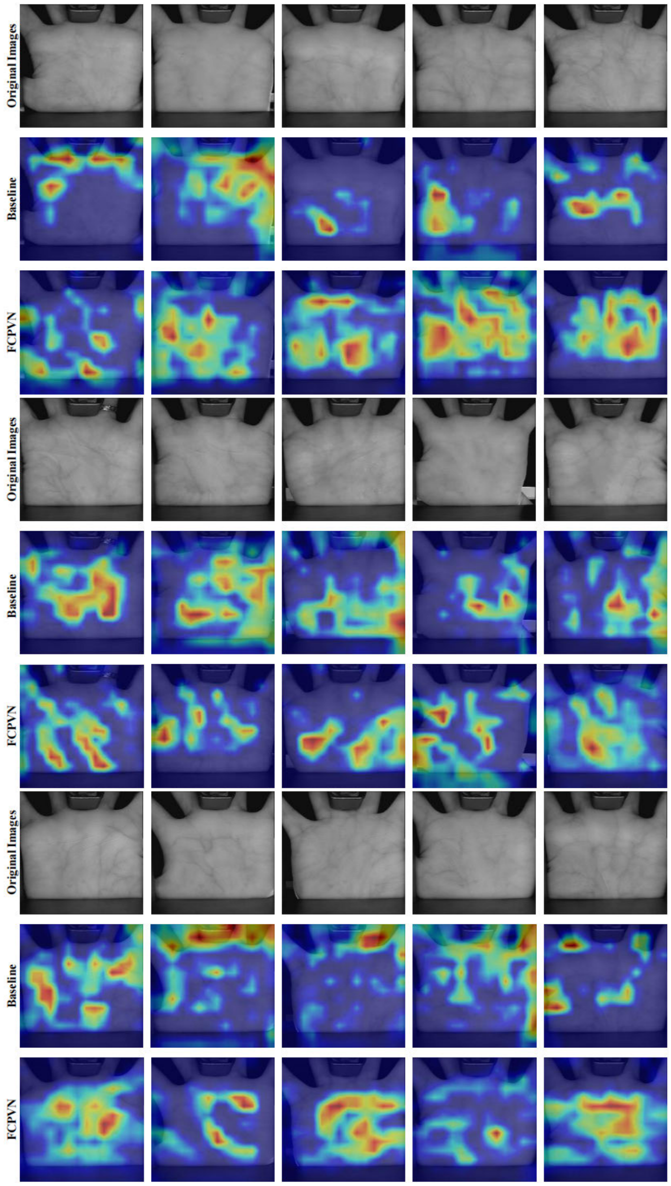


Fig. 14. Action maps visualization on the PolyU database. The text on the left of the images indicates the category of each image: original image, heatmap based on baseline, or heatmap based on FCPVN.

softmax function over all classes, resulting in exponential complexity growth with an increasing number of classes. The proposed method is more efficient and scalable for large-scale datasets with many classes due to its use of similarity information and constraints on feature distances between samples to learn more discriminative features.

Finally, in Section IV-G, we apply the algorithm in a real-world scenario and the experiment shows that the algorithm still meets the requirements for high accuracy and speed in practical applications.

V. Conclusion

This article proposes an effective network FCPVN for palm vein authentication based on contrastive learning. In addition, a novel loss function named focal contrastive loss is proposed, which makes the network narrow the intraclass features and push the interclass features apart. Considering that each example has a different learning difficulty, a hard example mining strategy is added to adjust the weight of each example involved in online training, which further improves the network performance.

The experimental results fully demonstrate the effectiveness of the proposed method compared to other methods. FCPVN is more concise and intuitive with a low EER in palm vein authentication tasks. The ablation studies are conducted to explore the contributions of each individual component of the algorithm and verify their necessity. In addition, the sensitivity of hyperparameters is determined, which can provide a reference for the practical application of the proposed method.

Our works have shown that contrastive learning has great potential for the palm vein authentication task. However, the proposed hard example mining strategy still has limited performance improvement with different palm vein databases. Therefore, how to classify the examples more carefully according to the difficulty in the database and how to design effective adaptive algorithms to adjust the contribution of each sample to model learning are the directions of the subsequent research.

Appendix

Feature Map Visualization

To avoid the precipice and subjective, we conducted visualization experiments on multiple samples for each dataset. This approach allowed us to obtain a more representative and robust evaluation of the models' performance, reducing the potential impact of individual samples and ensuring the reliability of our results. In the following, we present the visualization results of multiple samples for each dataset on both baseline and FCPVNet, providing a comprehensive and accurate assessment of the models' capabilities (see Figs. 10–14).

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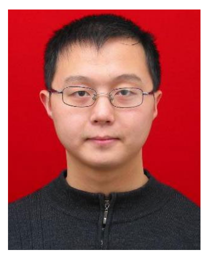

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